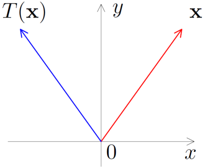
MATH1231 Mathematics 1B – Algebra

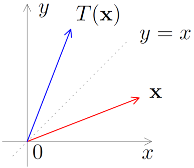
Section 10 - Eigenvalues and Eigenvectors

Lecturer: Aidan Sims (aidan.sims@unsw.edu.au)
Slides: David Angell, Laure Helme-Guizon and others

Investigation 1. i), interesting directions for reflections.

The following diagrams illustrate a reflection in the y–axis, and a reflection in the line $y = x$.





For each of these reflections, are there vectors $\overrightarrow{\mathbf{v}}$ whose image $T(\overrightarrow{\mathbf{v}})$ under the reflection is parallel to the original vector $\overrightarrow{\mathbf{v}}$?

In other words, are there vectors that satisfy $T(\overrightarrow{\mathbf{v}}) = \lambda \overrightarrow{\mathbf{v}}$ for some scalar λ?

Investigation 1. ii), Calculus.

Solve the differential equation

$$\frac{dy}{dx} = \lambda y \ .$$

The connection of this example with the previous ones is that differentiation and matrix multiplication are both linear transformations: the present problem can be written in the form

$$D(y) = \lambda y \ .$$

From calculus lectures you know that the equation has solutions for any real value of λ, and specifically that

$$y = Ce^{\lambda x} \ .$$

Investigation 1. iii), Powers of matrices.

1. Calculate D^2, D^3, D^4 and guess a formula for D^k, where D is the matrix

$$D = \begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix}.$$

2. Can you easily do the same for the matrix $A = \begin{pmatrix} 1 & -2 \\ -1 & 0 \end{pmatrix}$?

3. What if you are also told that $A = MDM^{-1}$, where $M = \begin{pmatrix} 2 & 1 \\ -1 & 1 \end{pmatrix}$?

The problem: Finding vectors parallel to their image.

Problem We are considering the equation $A\overrightarrow{\mathbf{v}} = \lambda\overrightarrow{\mathbf{v}}$, where A is a given matrix, λ is an unknown scalar and $\overrightarrow{\mathbf{v}}$ is an unknown vector. Note that

- the matrix A must be square;
- $\overrightarrow{\mathbf{v}} = \overrightarrow{\mathbf{0}}$ is always a solution, and so we are not really interested in this.

More generally, we consider $T(\overrightarrow{\mathbf{v}}) = \lambda\overrightarrow{\mathbf{v}}$, where T is a given linear transformation, λ is an unknown scalar and $\overrightarrow{\mathbf{v}}$ is an unknown vector. In this case

- $T(\overrightarrow{\mathbf{v}})$ is a scalar times $\overrightarrow{\mathbf{v}}$ and therefore must be in the same vector space as $\overrightarrow{\mathbf{v}}$: that is, T is a function from a vector space V to *the same* vector space V;
- as above, $\overrightarrow{\mathbf{v}} = \overrightarrow{\mathbf{0}}$ is always a solution and therefore is not of interest.

We give spacial names to the vectors parallel to their image and to the corresponding constants of proportionality.

Definition (Eigenvalues and eigenvectors of a linear map)

Let $T : V \to V$ be a linear transformation, λ a scalar and $\overrightarrow{\mathbf{v}}$ a non–zero vector in V. If

$$T(\overrightarrow{\mathbf{v}}) = \lambda \overrightarrow{\mathbf{v}}$$

then we say that λ is an **eigenvalue** of T, and that $\overrightarrow{\mathbf{v}}$ is an **eigenvector** of T **corresponding to** the eigenvalue λ.

Definition (Eigenvalues and eigenvectors of a square matrix)

Let A be an $n \times n$ matrix with real or complex entries, λ a complex number and $\overrightarrow{\mathbf{v}}$ a non–zero vector in $\mathbb{C}^n$. If

$$A\overrightarrow{\mathbf{v}} = \lambda \overrightarrow{\mathbf{v}}$$

then we say that λ is an **eigenvalue** of A, and that $\overrightarrow{\mathbf{v}}$ is an **eigenvector** of A **corresponding to** the eigenvalue λ.

Note that even if A is a real matrix we have to be prepared to find complex eigenvalues and eigenvectors: the reason for this will become clear later.

Example 2.

Consider the matrix A and the vectors $\overrightarrow{\mathbf{v_1}}$, $\overrightarrow{\mathbf{v_2}}$, $\overrightarrow{\mathbf{v_3}}$, given below:

$$A = \begin{pmatrix} 1 & -2 \\ -1 & 0 \end{pmatrix}, \quad \overrightarrow{\mathbf{v_1}} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \quad \overrightarrow{\mathbf{v_2}} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \text{and} \quad \overrightarrow{\mathbf{v_3}} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}.$$

Which of those vectors are eigenvectors of A? For the eigenvectors, what is the corresponding eigenvalue?

Solution.

- $A\overrightarrow{\mathbf{v_1}} = 2\overrightarrow{\mathbf{v_1}}$, so $\overrightarrow{\mathbf{v_1}}$ is an eigenvector associated to the eigenvalue $\lambda_1 = 2$.
- $A\overrightarrow{\mathbf{v_2}} = -\overrightarrow{\mathbf{v_2}}$, so $\overrightarrow{\mathbf{v_2}}$ is an eigenvector associated to the eigenvalue $\lambda_2 = -1$.
- $A\overrightarrow{\mathbf{v_3}}$ is not parallel to $\overrightarrow{\mathbf{v_3}}$, therefore $\overrightarrow{\mathbf{v_3}}$ is *not* an eigenvector of A.

Geometric significance of eigenvalues and eigenvectors, ctnd.

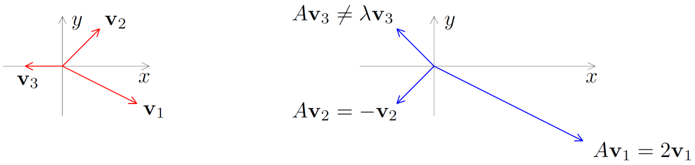
Example 2, illustrated. Recall that

$$A = \begin{pmatrix} 1 & -2 \\ -1 & 0 \end{pmatrix}, \quad \vec{\mathbf{v_1}} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \quad \vec{\mathbf{v_2}} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \text{and} \quad \vec{\mathbf{v_3}} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}.$$

and that we have shown that

- $A\vec{\mathbf{v_1}} = 2\vec{\mathbf{v_1}}$, so $\vec{\mathbf{v_1}}$ is an eigenvector associated to the eigenvalue $\lambda_1 = 2$.
- $A\vec{\mathbf{v_2}} = -\vec{\mathbf{v_2}}$, so $\vec{\mathbf{v_2}}$ is an eigenvector associated to the eigenvalue $\lambda_2 = -1$.
- $A\vec{\mathbf{v_3}}$ is not parallel to $\vec{\mathbf{v_3}}$, therefore $\vec{\mathbf{v_3}}$ is *not* an eigenvector of A.

What does that mean geometrically?



From the diagrams we see that the eigenvectors of A are those which still point in the same direction (or the opposite direction) if we multiply them by A, and the eigenvalues are the (signed) ratios of lengths between $A\vec{\mathbf{v}}$ and $\vec{\mathbf{v}}$.

Problem *But could we have found the eigenvalues and the eigenvectors if they had not been given to us?*

Computing eigenvalues and eigenvectors

This is how we find the eigenvalues and the eigenvectors in practice.

◇ The equation $A\overrightarrow{\mathbf{v}} = \lambda\overrightarrow{\mathbf{v}}$ can be rewritten

$$(A - \dots\lambda I)\overrightarrow{\mathbf{v}} = \overrightarrow{\mathbf{0}} \ , \qquad (*)$$

where I is the $n \times n$ identity matrix.

From the theory of linear equations and determinants studied in MATH1131 we see that this equation has a non–zero solution $\overrightarrow{\mathbf{v}}$ if and only if

$$\det(A - \lambda I) = 0 \ .$$

◇ This gives an equation, called the **characteristic equation** of A, which we can solve to find the eigenvalues λ;

◇ we then substitute these values one at a time into $(*)$ and solve to find the eigenvectors $\overrightarrow{\mathbf{v}}$ corresponding to each eigenvalue.

Example 3. *Same matrix as in example (ii) in the investigation and in Example 2.*

Find the eigenvalues and the eigenvectors of the matrix $A = \begin{pmatrix} 1 & -2 \\ -1 & 0 \end{pmatrix}$.

Worked solution.

- First, we find the eigenvalues. We have

$$\det(A - \lambda I) = \det \begin{pmatrix} 1-\lambda & -2 \\ -1 & -\lambda \end{pmatrix} = (1-\lambda)(-\lambda) - 2 = \lambda^2 - \lambda - 2 \ ;$$

  solving the characteristic equation $\lambda^2 - \lambda - 2 = 0$ gives eigenvalues $\lambda = 2$ and $\lambda = -1$.

- Then, to find the eigenvectors, we substitute these values of λ into $(A - \lambda I)\overrightarrow{\mathbf{v}} = \overrightarrow{\mathbf{0}}\,(*)$ and solve for $\overrightarrow{\mathbf{v}}$.
  - For $\lambda = 2$ we have $(A - 2I)\overrightarrow{\mathbf{v}} = \overrightarrow{\mathbf{0}}$; solving by row–reduction and back substitution, we have

$$\left(\begin{array}{cc|c} -1 & -2 & 0 \\ -1 & -2 & 0 \end{array}\right) \longleftrightarrow \left(\begin{array}{cc|c} -1 & -2 & 0 \\ 0 & 0 & 0 \end{array}\right) \quad \text{so} \quad v_2 = t \ , \ v_1 = -2t \ .$$

    Hence, the eigenvectors for $\lambda = 2$ are $\overrightarrow{\mathbf{v}} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} -2t \\ t \end{pmatrix} = t\begin{pmatrix} -2 \\ 1 \end{pmatrix}$.
  - For $\lambda = -1$ we have $(A + I)\overrightarrow{\mathbf{v}} = \overrightarrow{\mathbf{0}}$ and

$$\left(\begin{array}{cc|c} 2 & -2 & 0 \\ -1 & 1 & 0 \end{array}\right) \longleftrightarrow \left(\begin{array}{cc|c} 1 & -1 & 0 \\ 0 & 0 & 0 \end{array}\right) \quad \text{so} \quad v_2 = s \ , \ v_1 = s \ .$$

    Hence, the eigenvectors for $\lambda = -1$ are $\overrightarrow{\mathbf{v}} = \begin{pmatrix} s \\ s \end{pmatrix} = s\begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

Ways to check your work

There is a fair deal of calculation involved in finding eigenvalues and eigenvectors, and it is easy to make arithmetic slips. Therefore it is very important to **check your work**.

On this page we give some checks and apply them to the first example: in future work you should check as you go along.

- For any matrix, the sum of the eigenvalues equals the trace of the matrix, that is, the sum of the diagonal elements from top left to bottom right. In the example above, the sum of the eigenvalues is $2+(-1)$, the trace is $1+0$, these are equal, so the check is OK.
  - When adding up the eigenvalues, if the same eigenvalue occurs more than once, then it must be added up more than once.
- When finding eigenvectors, there must always be an infinite number of solutions. In particular, if you ever find that you get the unique solution $\overrightarrow{\mathbf{v}} = \overrightarrow{\mathbf{0}}$ for an eigenvector, then you know for sure that you are wrong – go back, find the error and fix it. (*Hint*: the most likely error is that your eigenvalues are wrong.)
  - On the previous page, the echelon form for $A-2I$ has a non–leading column, so there are infinitely many eigenvectors and the check holds. The same goes for $A+I$.
- Finally, once you have found λ and $\overrightarrow{\mathbf{v}}$, check that $A\overrightarrow{\mathbf{v}} = \lambda\overrightarrow{\mathbf{v}}$. In example 1 we have

$$A\begin{pmatrix}-2\\1\end{pmatrix} = \begin{pmatrix}-4\\2\end{pmatrix} = 2\begin{pmatrix}-2\\1\end{pmatrix} \quad \text{and} \quad A\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}-1\\-1\end{pmatrix} = -\begin{pmatrix}1\\1\end{pmatrix}\ .$$

Using these checks on an example.

Example 4. Let $B = \begin{pmatrix} 1 & 2 & -1 \\ -6 & -7 & 3 \\ -4 & -4 & 1 \end{pmatrix}$.

Find its eigenvalues and its eigenvectors, and use the checks described on the previous slide to check your work.

Solution.

$$\det(B - \lambda I) = \det \begin{pmatrix} 1-\lambda & 2 & -1 \\ -6 & -7-\lambda & 3 \\ -4 & -4 & 1-\lambda \end{pmatrix} = \cdots = -(\lambda+1)^2(\lambda+3) \, ,$$

so the eigenvalues of B are -1 and -3. We say that -1 has **algebraic multiplicity** 2 while -3 has algebraic multiplicity 1.

Check:

- the trace of the matrix is $\ldots 1 + (-7) + 1$
- the sum of the eigenvalues is $\ldots (-1) + (-1) + (-3)$,
- and these $\ldots$ are equal, so our eigenvalues could be right.

Note that the eigenvalue -1 is counted twice in the sum the eigenvalues because its algebraic multiplicity is 2. If it algebraic multiplicity had been 5, it would had been counted five times...etc.

Using these checks on an example.

Example 4, continued.

Calculating eigenvectors, for $\lambda = -1$ we have

$$B - \lambda I = \begin{pmatrix} 2 & 2 & -1 \\ -6 & -6 & 3 \\ -4 & -4 & 2 \end{pmatrix} \longleftrightarrow \begin{pmatrix} 2 & 2 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} .$$

Check: there are infinitely many solutions.

Solving gives $v_3 = t$, $v_2 = s$, $v_1 = -s + \frac{1}{2}t$, and the eigenvectors for $\lambda = -1$ are

$$\overrightarrow{\mathbf{v}} = \begin{pmatrix} -s + \frac{1}{2}t \\ s \\ t \end{pmatrix} = s \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} \frac{1}{2} \\ 0 \\ 1 \end{pmatrix} .$$

Check: we have

$$B \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = - \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} \quad \text{and} \quad B \begin{pmatrix} \frac{1}{2} \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} \\ 0 \\ -1 \end{pmatrix} = - \begin{pmatrix} \frac{1}{2} \\ 0 \\ 1 \end{pmatrix} .$$

so these are indeed eigenvectors for $\lambda = -1$

Using these checks on an example.

Example 4, continued.

For $\lambda = -3$ we obtain

$$B - \lambda I = \begin{pmatrix} 4 & 2 & -1 \\ -6 & -4 & 3 \\ -4 & -4 & 4 \end{pmatrix} \longleftrightarrow \begin{pmatrix} 4 & 2 & -1 \\ 0 & -2 & 3 \\ 0 & 0 & 0 \end{pmatrix}$$

(**check**: infinitely many solutions), which gives eigenvectors

$$\overrightarrow{\mathbf{v}} = t \begin{pmatrix} -\frac{1}{2} \\ \frac{3}{2} \\ 1 \end{pmatrix} = \frac{1}{2} t \begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix} = u \begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix} .$$

Final **check**: $B \begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ -9 \\ -6 \end{pmatrix} = -3 \begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix}$.

Fundamental results

Here are the core results on existence and computation of eigenvalues and eigenvectors.

Fact (Computing eigenvalues and eigenvectors)

Let A be an $n \times n$ real or complex matrix, λ a real or complex scalar and $\overrightarrow{\mathbf{v}}$ a non–zero vector in $\mathbb{C}^n$. Then

- λ is an eigenvalue of A if and only if $\det(A - \lambda I) = 0$;
- $\overrightarrow{\mathbf{v}}$ is an eigenvector of A corresponding to the eigenvalue λ if and only if $(A - \lambda I)\overrightarrow{\mathbf{v}} = \overrightarrow{\mathbf{0}}$.

Fact (The characteristic polynomial)

Let A be an $n \times n$ matrix. Then $\det(A - \lambda I)$ is a polynomial of degree n in λ.

Fact (Existence of eigenvalues)

An $n \times n$ real or complex matrix has exactly n eigenvalues, provided that

- both real and complex eigenvalues are counted;
- eigenvalues are counted "according to their multiplicity".

Proof. Under the stated conditions, an nth degree polynomial has exactly n roots.

Comments

- Given an $n \times n$ matrix A and an eigenvalue λ, we refer to the set

  $$E_\lambda = \{ \overrightarrow{\mathbf{v}} \in \mathbb{C}^n \mid A\overrightarrow{\mathbf{v}} = \lambda \overrightarrow{\mathbf{v}} \}$$

  as the **eigenspace** of A corresponding to λ. To prove that E_λ is a subspace of $\mathbb{C}^n$, use the Subspace Theorem (*exercise*); alternatively, we have $E_\lambda = \ker(A - \lambda I)$, and we know from Chapter 7 that the kernel of a matrix is always a vector space.
  - If λ is not an eigenvalue of A then $E_\lambda = \left\{ \overrightarrow{\mathbf{0}} \right\}$; if λ is an eigenvalue then E_λ is more than $\left\{ \overrightarrow{\mathbf{0}} \right\}$.

- The determinant of $A - \lambda I$ is called the **characteristic polynomial** of A.
  - Note that if a matrix A has real entries then its characteristic polynomial will have real coefficients, but it may still have complex roots. So a real matrix may have complex eigenvalues.
  - Observe on page 12 that B has an eigenvalue $\lambda = -1$ with algebraic multiplicity 2, and that there are two linearly independent eigenvectors corresponding to this eigenvalue. In fact, it is always true that

    $$\left\{ \begin{array}{c} \text{number of linearly} \\ \text{independent eigenvectors} \end{array} \right\} \leqslant \left\{ \begin{array}{c} \text{algebraic multiplicity} \\ \text{of the eigenvalue} \end{array} \right\},$$

    but it is not always true that these quantities are equal.

A matrix with real entries but complex eigenvalues

For the next two examples, please fill in the working we have omitted, and check the working as you go.

Example 5. For $A = \begin{pmatrix} 1 & 2 \\ -1 & -1 \end{pmatrix}$ we have

$$\det(A - \lambda I) = \det \begin{pmatrix} 1-\lambda & 2 \\ -1 & -1-\lambda \end{pmatrix} = \lambda^2 + 1 \ ,$$

so A has eigenvalues $\lambda = \pm i$. Finding eigenvectors for $\lambda = i$, we have

$$A - \lambda I = \begin{pmatrix} 1-i & 2 \\ -1 & -1-i \end{pmatrix} \longleftrightarrow \begin{pmatrix} 1 & 1+i \\ 0 & 0 \end{pmatrix} \qquad \text{so} \qquad \overrightarrow{\mathbf{v}} = t \begin{pmatrix} 1+i \\ -1 \end{pmatrix} \ .$$

Similarly, the eigenvectors for $\lambda = -i$ are

$$\overrightarrow{\mathbf{v}} = t \begin{pmatrix} 1-i \\ -1 \end{pmatrix} \ .$$

Note that A is a real matrix (meaning a matrix with real entries), but its eigenvalues and eigenvectors are complex.

Eigenvalues with algebraic multiplicity greater than 1

Example 6. For $C = \begin{pmatrix} 1 & 1 \\ -9 & 7 \end{pmatrix}$ we have

$$\det(C - \lambda I) = \det \begin{pmatrix} 1-\lambda & 1 \\ -9 & 7-\lambda \end{pmatrix} = (\lambda - 4)^2 \ .$$

Thus C has one eigenvalue $\lambda = 4$, with algebraic multiplicity 2. From

$$C - 4I = \begin{pmatrix} -3 & 1 \\ -9 & 3 \end{pmatrix} \longleftrightarrow \begin{pmatrix} -3 & 1 \\ 0 & 0 \end{pmatrix}$$

we see that the corresponding eigenvectors are $\overrightarrow{\mathbf{v}} = t \begin{pmatrix} 1 \\ 3 \end{pmatrix}$.

Note that here we have an eigenvalue of algebraic multiplicity 2, but it is impossible to find two linearly independent eigenvectors – contrast the example on page 12.

For eigenvalues with algebraic multiplicity greater than 1, the dimension of the eigenspace can be strictly less than the algebraic multiplicity of the eigenvalue (they could also be equal, like in the example on page 12).

Challenge problem. Let A be a 2×2 matrix with an eigenvalue of multiplicity 2 for which there are two independent eigenvectors. Prove that A is a scalar times the identity matrix.

Example 7.

Recall that in the investigation and in example 2 on page 6 we looked at the matrix

$$A = \begin{pmatrix} 1 & -2 \\ -1 & 0 \end{pmatrix}$$

which has eigenvalues and eigenvectors

$$\lambda_1 = 2 \ , \ \overrightarrow{\mathbf{v_1}} = \begin{pmatrix} 2 \\ -1 \end{pmatrix} \quad \text{and} \quad \lambda_2 = -1 \ , \ \overrightarrow{\mathbf{v_2}} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \ ;$$

we also chose

$$\overrightarrow{\mathbf{v_3}} = \begin{pmatrix} -1 \\ 0 \end{pmatrix} \ .$$

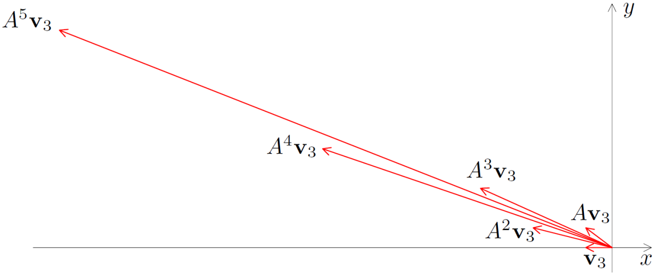
What happens if we repeatedly multiply $\overrightarrow{\mathbf{v_3}}$ by A? We have

$$A\overrightarrow{\mathbf{v_3}} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \ A^2\overrightarrow{\mathbf{v_3}} = \begin{pmatrix} -3 \\ 1 \end{pmatrix}, \ A^3\overrightarrow{\mathbf{v_3}} = \begin{pmatrix} -5 \\ 3 \end{pmatrix}, \ A^4\overrightarrow{\mathbf{v_3}} = \begin{pmatrix} -11 \\ 5 \end{pmatrix}, \ A^5\overrightarrow{\mathbf{v_3}} = \begin{pmatrix} -21 \\ 11 \end{pmatrix}$$

Let's see what this looks like on a diagram.

....continued

Example 7, continued. From the diagram,



the vectors $A^k\overrightarrow{\mathbf{v}_3}$ seem to be increasing in length and settling down towards a fixed direction, but beyond this it is hard to say exactly what is going on.
What if we do the same thing starting with $\overrightarrow{\mathbf{v}_1}$? This is easy! We have $A\overrightarrow{\mathbf{v}_1} = 2\overrightarrow{\mathbf{v}_1}$, so

$$A^2\overrightarrow{\mathbf{v}_1} = 2^2\overrightarrow{\mathbf{v}_1}\ , \quad A^3\overrightarrow{\mathbf{v}_1} = 2^3\overrightarrow{\mathbf{v}_1}\ , \ldots , \quad A^k\overrightarrow{\mathbf{v}_1} = 2^k\overrightarrow{\mathbf{v}_1}\ ;$$

the vectors are all in the same direction, and they double in length at each step.
Similarly,

$$A^k\overrightarrow{\mathbf{v}_2} = (-1)^k\overrightarrow{\mathbf{v}_2}\ .$$

....continued

Powers of matrices, continued

Example 7, continued. The results we have just found for $\overrightarrow{\mathbf{v_1}}$ and $\overrightarrow{\mathbf{v_2}}$ enable us to settle the corresponding question for $\overrightarrow{\mathbf{v_3}}$. The set $\{\, \overrightarrow{\mathbf{v_1}}, \overrightarrow{\mathbf{v_2}} \,\}$ is linearly independent (why?) and is therefore a basis for $\mathbb{R}^2$ (why?), so we can write $\overrightarrow{\mathbf{v_3}}$ as a linear combination of $\overrightarrow{\mathbf{v_1}}$ and $\overrightarrow{\mathbf{v_2}}$. You can check for yourself that in fact

$$\overrightarrow{\mathbf{v_3}} = -\frac{1}{3}\overrightarrow{\mathbf{v_1}} - \frac{1}{3}\overrightarrow{\mathbf{v_2}} \ ;$$

therefore

$$A^k\overrightarrow{\mathbf{v_3}} = -\frac{1}{3}(A^k\overrightarrow{\mathbf{v_1}} + A^k\overrightarrow{\mathbf{v_2}}) = -\frac{1}{3}(2^k\overrightarrow{\mathbf{v_1}} + (-1)^k\overrightarrow{\mathbf{v_2}}) \ ,$$

and we have obtained a formula for $A^k\overrightarrow{\mathbf{v_3}}$ in which we do not need to evaluate powers of matrices. We can also write

$$A^k\overrightarrow{\mathbf{v_3}} = -\frac{1}{3}\,2^k\left(\overrightarrow{\mathbf{v_1}} + \left(-\frac{1}{2}\right)^k\overrightarrow{\mathbf{v_2}}\right) \ ,$$

which is approximately $-\frac{1}{3}\,2^k\overrightarrow{\mathbf{v_1}}$ if k is large. Hence, for large k, the magnitude of $A^k\overrightarrow{\mathbf{v_3}}$ is approximately $\frac{1}{3}\,2^k|\overrightarrow{\mathbf{v_1}}|$, and its direction is close to that of $-\overrightarrow{\mathbf{v_1}}$. *Check* these statements by looking at the diagram on the previous page.

Powers and diagonal matrices

The preceding exercise suggests that problems involving powers of matrices may be simplified by using properties of eigenvalues and eigenvectors.

In fact, notice that

$$A(\,\overrightarrow{\mathbf{v_1}}\mid\overrightarrow{\mathbf{v_2}}\,) = (\,A\overrightarrow{\mathbf{v_1}}\mid A\overrightarrow{\mathbf{v_2}}\,) = (\,2\overrightarrow{\mathbf{v_1}}\mid -\overrightarrow{\mathbf{v_2}}\,) = (\,\overrightarrow{\mathbf{v_1}}\mid\overrightarrow{\mathbf{v_2}}\,)\begin{pmatrix}2 & 0\\ 0 & -1\end{pmatrix}\ ,$$

or

$$AM = MD\ .$$

Here A is the given matrix, D is a matrix with the eigenvalues of A along the diagonal and zeros elsewhere, and M is a matrix whose columns are the eigenvectors of A. In this case the eigenvectors are linearly independent, so M is invertible and we can **diagonalise** the matrix A by writing

$$A = MDM^{-1}\ .$$

We can now without much difficulty evaluate powers of A. We have

$$A^k = \overbrace{(MDM^{-1})(MDM^{-1})(MDM^{-1})\cdots(MDM^{-1})}^{k\ \text{times}} = MD^kM^{-1}\ ;$$

the matrices M and D are known, M^{-1} can be found, and it is easy to calculate D^k.

Calculating powers of matrices using diagonalisation

This example shows how to calculate powers of matrices by using the method of diagonalisation.

Example 8(a). Diagonalise $A = \begin{pmatrix} 4 & -2 \\ -6 & 5 \end{pmatrix}$ and find a formula for A^k.

Solution. By standard methods we find the eigenvalues and eigenvectors of A,

$$\lambda = 1 \, , \ \vec{\mathbf{v}} = t \begin{pmatrix} 2 \\ 3 \end{pmatrix} \quad \text{and} \quad \lambda = 8 \, , \ \vec{\mathbf{v}} = t \begin{pmatrix} 1 \\ -2 \end{pmatrix} \ ;$$

therefore, $A = MDM^{-1}$ with

$$D = \begin{pmatrix} 1 & 0 \\ 0 & 8 \end{pmatrix} \quad \text{and} \quad M = \begin{pmatrix} 2 & 1 \\ 3 & -2 \end{pmatrix} \ .$$

***Food for thought**. Are there any alternative choices for M and D?*

From the diagonalisation we have

$$A^k = MD^kM^{-1} = -\frac{1}{7}\begin{pmatrix} 2 & 1 \\ 3 & -2 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 8^k \end{pmatrix}\begin{pmatrix} -2 & -1 \\ -3 & 2 \end{pmatrix} = \frac{1}{7}\begin{pmatrix} 4+3\times 8^k & 2-2\times 8^k \\ 6-6\times 8^k & 3+4\times 8^k \end{pmatrix}.$$

Having finished this kind of problem you should always **check** that your answer is correct for $k = 0$ and for $k = 1$.

Calculating powers of matrices using diagonalisation

With a 3×3 matrix!

Example 8(b). Do the same for $B = \begin{pmatrix} 1 & 2 & -1 \\ -6 & -7 & 3 \\ -4 & -4 & 1 \end{pmatrix}$. i.e. diagonalise B and find a formula for B^k.

Solution. From Example 4 pages 11–13 we know that B has eigenvalues and eigenvectors

$$\lambda = -1\ ,\ \vec{\mathbf{v}} = s\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + t\begin{pmatrix} \frac{1}{2} \\ 0 \\ 1 \end{pmatrix} \quad \text{and} \quad \lambda = -3\ ,\ \vec{\mathbf{v}} = u\begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix}\ .$$

Three linearly independent eigenvectors can be found, and so B is diagonalisable with, for example,

$$D = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -3 \end{pmatrix} \quad \text{and} \quad M = \begin{pmatrix} -1 & 1 & -1 \\ 1 & 0 & 3 \\ 0 & 2 & 2 \end{pmatrix}\ .$$

Therefore

$$B^k = MD^kM^{-1}\ .$$

Exercise. Complete the calculation to find an explicit formula for B^k.

A non–diagonalisable matrix

Will this great idea work every time? Let's try with another matrix!

Example 8(c). Do the same for $C = \begin{pmatrix} 1 & 1 \\ -9 & 7 \end{pmatrix}$ i.e. diagonalise C and find a formula for C^k.

Solution. From Example 6 page 17, the matrix C has the eigenvalue $\lambda = 4$ only, with multiplicity 2, and eigenvectors

$$\vec{\mathbf{v}} = t\begin{pmatrix} 1 \\ 3 \end{pmatrix} .$$

Therefore C does not have two *linearly independent* eigenvectors, it is not diagonalisable, and we cannot find powers of C by this method.

Yes, bad news, some matrices are not diagonalisable!

Consolation price: $C = MJM^{-1}$ where

$$J = \begin{pmatrix} 4 & 1 \\ 0 & 4 \end{pmatrix} \quad \text{and} \quad M = \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix} .$$

The matrix J is called a **Jordan form** of C and is "the next best thing" when C is not diagonalisable.

Challenge. See if you can use this information to find a formula for C^k.

Criteria for diagonalisation

We have seen examples of matrices which are and which are not diagonalisable. We now investigate the properties of eigenvalues and eigenvectors which make a matrix diagonalisable.

First we note that we have succeeded in writing an $n \times n$ matrix in the form $A = MDM^{-1}$ by taking the columns of M to be eigenvectors of A; in order for M to be invertible, it is necessary that these eigenvectors be linearly independent. Here is our first result.

Fact (Diagonalisation of matrices)

An $n \times n$ matrix A is diagonalisable if and only if it has n linearly independent eigenvectors.

Proof. Difficult – not examinable for MATH1231. However, the argument essentially follows the procedure in the example on page 21; for more details, see the course notes.

Fact (Diagonalisation procedure)

Suppose that A is an $n \times n$ matrix with n linearly independent eigenvectors. Then $A = MDM^{-1}$, where M is a matrix whose columns consist of linearly independent eigenvectors of A, and D is the diagonal matrix whose kth diagonal entry is the eigenvalue corresponding to the kth column of M.

Proof. Again, look at the example on page 21 to get an idea of how the proof would go.

Independence of eigenvectors

Observe that the only case we have seen (C on pages 17 and 25) in which an $n \times n$ matrix does not have n independent eigenvectors is a case where there is a repeated eigenvalue.

Fact (Independence of eigenvectors)

Let $\vec{\mathbf{v}}_1, \dots, \vec{\mathbf{v}}_k$ be eigenvectors of a matrix A. If the corresponding eigenvalues are all different, then the eigenvectors are linearly independent.

Proof. Omitted.

Corollary (Matrices without repeated eigenvalues are diagonalisable)

If A is an $n \times n$ matrix with n different eigenvalues, then A is diagonalisable.

Proof. If A has n different eigenvalues, choose an eigenvector for each eigenvalue. By the theorem we have just proved, these eigenvectors are independent; and by the theorem before that, A is diagonalisable.

The converse of this corollary and that of the previous theorem are *not true*:

- an $n \times n$ matrix which has repeated eigenvalues (and therefore does not have n different eigenvalues) **may** be diagonalisable;
- eigenvectors which belong to the same eigenvalue **may** be linearly independent.

  For an example of both of these, recall the matrix B from page 24.

Another example

Example 9. Find a formula for A^k, where $A = \begin{pmatrix} 1 & 3 \\ 4 & 5 \end{pmatrix}$.

Solution. The characteristic polynomial of A is

$$det(A - \lambda I) = \det \begin{pmatrix} 1-\lambda & 3 \\ 4 & 5-\lambda \end{pmatrix}$$

$$= (1-\lambda)(5-\lambda) - 12 = \lambda^2 - 6\lambda - 7 = (\lambda - 7)(\lambda + 1),$$

and the eigenvalues of A are the roots of this polynomial: $\lambda = 7$ and $\lambda = -1$.
Finding the eigenvectors for $\lambda = 7$, we have

$$A - 7I = \begin{pmatrix} -6 & 3 \\ 4 & -2 \end{pmatrix} \longleftrightarrow \begin{pmatrix} 2 & -1 \\ 0 & 0 \end{pmatrix} , \quad \text{so} \quad \vec{\mathbf{v}} = t \begin{pmatrix} 1 \\ 2 \end{pmatrix} .$$

For $\lambda = -1$ we have

$$A + I = \begin{pmatrix} 2 & 3 \\ 4 & 6 \end{pmatrix} \longleftrightarrow \begin{pmatrix} 2 & 3 \\ 0 & 0 \end{pmatrix} , \quad \text{so} \quad \vec{\mathbf{v}} = t \begin{pmatrix} -3 \\ 2 \end{pmatrix} .$$

....continued

Example 9, continued. We have shown that

$$\vec{\mathbf{v}} = t\begin{pmatrix}1\\2\end{pmatrix} \text{ is an eigenvector for } \lambda = 7 \text{ and}$$

$$\vec{\mathbf{v}} = t\begin{pmatrix}-3\\2\end{pmatrix} \text{ is an eigenvector for } \lambda = -1.$$

Hence A can be diagonalised as $A = MDM^{-1}$, where (among other possibilities)

$$D = \begin{pmatrix}7 & 0\\0 & -1\end{pmatrix}, \qquad M = \begin{pmatrix}1 & -3\\2 & 2\end{pmatrix}.$$

Therefore

$$\begin{aligned} A^k = MD^kM^{-1} &= \frac{1}{8}\begin{pmatrix}1 & -3\\2 & 2\end{pmatrix}\begin{pmatrix}7^k & 0\\0 & (-1)^k\end{pmatrix}\begin{pmatrix}2 & 3\\-2 & 1\end{pmatrix} \\ &= \frac{1}{8}\begin{pmatrix}2\times 7^k + 6(-1)^k & 3\times 7^k - 3(-1)^k\\4\times 7^k - 4(-1)^k & 6\times 7^k + 2(-1)^k\end{pmatrix}. \end{aligned}$$

Check that when $k = 0$ this formula gives us I, and when $k = 1$ it gives us the original matrix A.

Kahoot on Eigenvalues, Eigenvectors and Diagonalisation

Definition

Let $T : V \to V$ be a linear transformation, λ a scalar and $\overrightarrow{\mathbf{v}}$ a non–zero vector in V. If $T(\overrightarrow{\mathbf{v}}) = \lambda \overrightarrow{\mathbf{v}}$, then we say that λ is an **eigenvalue** of T, and that $\overrightarrow{\mathbf{v}}$ is an **eigenvector** of T **corresponding to** the eigenvalue λ.

Fact (Diagonalisation procedure)

Suppose that A is an $n \times n$ matrix with n linearly independent eigenvectors. Then $A = MDM^{-1}$, where M is a matrix whose columns consist of linearly independent eigenvectors of A, and D is the diagonal matrix whose kth diagonal entry is the eigenvalue corresponding to the kth column of M.

- Students use https://kahoot.it/ or the Kahoot app on their phone.
- The lecturer starts the game using:
https://create.kahoot.it/share/math1231-alg-eigenvalues-eigenvectors/d277c28d-502d-4759-b9cb-ed16115b2c8e
or
Kahoot: MATH1231 - Eigenvalues, Eigenvectors and Diagonalisation

Diagonalisation and systems of differential equations

- We have seen several examples of how matrix diagonalisation can help calculate powers of matrices.

- More surprisingly, matrix diagonalisation can help solve system of 'coupled' differential equations ('coupled' in the sense that it is not obvious how to separate your ODEs into an ODE about $y_1(t)$ and an ODE about $y_2(t)$.

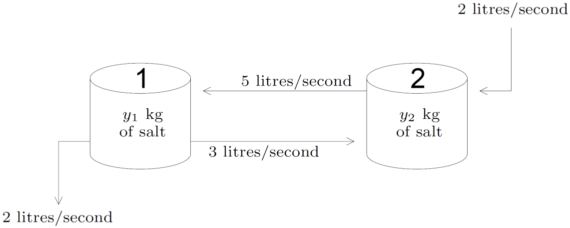
  You do not need matrices to find solutions to the system:
  $$\begin{cases} \frac{dy_1}{dt} = 3y_1 \\ \frac{dy_2}{dt} = 4y_2, \end{cases}$$

  but what about the following system?
  $$\begin{cases} \frac{dy_1}{dt} = 6y_1 + 2y_2 \\ \frac{dy_2}{dt} = -2y_1 + y_2 \end{cases}$$

Here is a question similar to some you have seen in calculus.

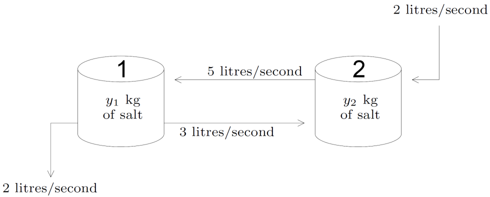
Example 10. A tank contains 100 litres of water with initially 4 kilograms of salt dissolved in it; another tank contains 100 litres of water with initially 7 kilograms of salt. The liquid in each tank is constantly stirred. Liquid is pumped from the first tank to the second at a rate of 3 litres/second, and from the second to the first at a rate of 5 litres/second. To keep the volumes constant, pure water is added to the second tank (2 litres/second) and salt solution is drained from the first (2 litres/second). Find the amount of salt in each tank at all future times.

Solution. At time t seconds, let there be y_1 kilograms of salt in the first tank and y_2 kilograms in the second.



....continued

Now consider a short time interval of length Δt, and let $\Delta y_1, \Delta y_2$ be the increases in y_1, y_2 over this interval.



- The outflow from tank 1 to tank 2 in time Δt is $3\Delta t$ litres. Since the mixture in tank 1 is thoroughly stirred, the *proportion* of salt in the outflow is the same as in the whole tank. So

$$\frac{\text{amount of salt lost}}{3\Delta t} = \frac{\text{amount in tank}}{100} = \frac{y_1}{100} ,$$

  approximately, and $0.03y_1\Delta t$ kilograms of salt is lost.
- For similar reasons, the inflow from tank 2 brings with it $0.05y_2\Delta t$ kilograms of salt. . .
- . . . and the liquid drained from tank 1 removes $0.02y_1\Delta t$ kilograms.

Hence

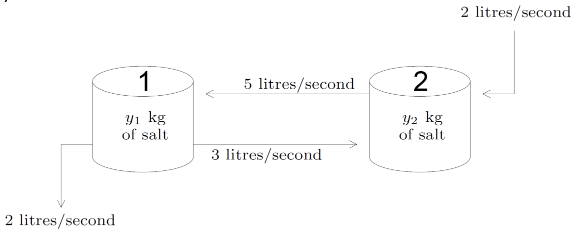
$$\Delta y_1 = -0.03y_1\Delta t + 0.05y_2\Delta t - 0.02y_1\Delta t ;$$

dividing by Δt, taking the limit as $\Delta t \to 0$, and doing the same sort of thing for y_2 yields a system of two simultaneous differential equations

$$\frac{dy_1}{dt} = -0.05y_1 + 0.05y_2 , \qquad \frac{dy_2}{dt} = 0.03y_1 - 0.05y_2 .$$

We also have the initial conditions $y_1(0) = 4$ and $y_2(0) = 7$.

◇ Summary:



The situation can be modelled by:

$$\begin{cases} \frac{dy_1}{dt} = -0.05y_1 + 0.05y_2 \\ \frac{dy_2}{dt} = 0.03y_1 - 0.05y_2 \end{cases}$$

with the initial conditions $y_1(0) = 4$ and $y_2(0) = 7$.

◇ If we write

$$\vec{\mathbf{y}} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \quad \text{and} \quad A = \begin{pmatrix} -0.05 & 0.05 \\ 0.03 & -0.05 \end{pmatrix},$$

then the above system of differential equations can be written as a single equation involving vectors and matrices,

$$\frac{d\vec{\mathbf{y}}}{dt} = A\vec{\mathbf{y}}. \qquad (*)$$

$$\text{With } \vec{\mathbf{y}} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \quad \text{and} \quad A = \begin{pmatrix} -0.05 & 0.05 \\ 0.03 & -0.05 \end{pmatrix}, \quad \frac{d\,\vec{\mathbf{y}}}{dt} = A\,\vec{\mathbf{y}}\ . \qquad (*)$$

By analogy with the *scalar* differential equation

$$\frac{dy}{dt} = ay\ ,$$

which has solutions $y = Ce^{at}$ where C is a constant, we guess that $(*)$ might have a solution $\vec{\mathbf{y}} = e^{\lambda t}\vec{\mathbf{v}}$ where $\vec{\mathbf{v}}$ is a (constant) vector.

◇ To see whether there exists solutions of this form, we substitute $\vec{\mathbf{y}} = e^{\lambda t}\vec{\mathbf{v}}$ into $(*)$:

$$\frac{d\,\vec{\mathbf{y}}}{dt} = A\,\vec{\mathbf{y}} \qquad \Leftrightarrow \qquad \lambda e^{\lambda t}\vec{\mathbf{v}} = Ae^{\lambda t}\vec{\mathbf{v}} \qquad \Leftrightarrow \qquad A\,\vec{\mathbf{v}} = \lambda\,\vec{\mathbf{v}}\ .$$

This argument proves the following result.

Fact (Solution of a system of differential equations)

The vector function $\vec{\mathbf{y}} = e^{\lambda t}\vec{\mathbf{v}}$ is a non–zero solution of $(*)$ if and only if λ is an eigenvalue of A and $\vec{\mathbf{v}}$ is an eigenvector of A corresponding to λ.

We will return to the salt–in–tanks problem at the very end.

Example 11. Solve the system

$$\frac{dy_1}{dt} = 6y_1 + 2y_2 \,, \qquad \frac{dy_2}{dt} = -2y_1 + y_2$$

subject to the initial conditions $y_1(0) = -5$ and $y_2(0) = 7$.

Worked solution. The system can be written

$$\frac{d\,\overrightarrow{\mathbf{y}}}{dt} = A\,\overrightarrow{\mathbf{y}} \quad \text{where} \quad \overrightarrow{\mathbf{y}} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \quad \text{and} \quad A = \begin{pmatrix} 6 & 2 \\ -2 & 1 \end{pmatrix} .$$

Now (*exercise!*) A has eigenvalues and eigenvectors

$$\lambda = 2\,, \ \overrightarrow{\mathbf{v}} = \alpha \begin{pmatrix} 1 \\ -2 \end{pmatrix} \quad \text{and} \quad \lambda = 5\,, \ \overrightarrow{\mathbf{v}} = \alpha \begin{pmatrix} 2 \\ -1 \end{pmatrix} \,,$$

so the system has solutions

$$\overrightarrow{\mathbf{y}} = e^{2t} \begin{pmatrix} 1 \\ -2 \end{pmatrix} \quad \text{and} \quad \overrightarrow{\mathbf{y}} = e^{5t} \begin{pmatrix} 2 \\ -1 \end{pmatrix} .$$

....continued

Example 11, continued.

As in calculus, the **general solution** of the system is

$$\overrightarrow{\mathbf{y}} = Ce^{2t}\begin{pmatrix}1\\-2\end{pmatrix} + De^{5t}\begin{pmatrix}2\\-1\end{pmatrix}$$

where C, D are constants, or in scalar form

$$y_1 = Ce^{2t} + 2De^{5t}\ , \qquad y_2 = -2Ce^{2t} - De^{5t}\ .$$

Note that this solution contains two arbitrary constants, not four – the two Cs really are the same, as are the two Ds. The functions y_1 and y_2 are not independent of each other.

◇ Now substituting $t = 0$ and using the given initial conditions, we have

$$C + 2D = -5\ , \quad -2C - D = 7\ .$$

Solving these equations yields $C = -3$, $D = -1$; hence, the solution of the initial value problem is

$$y_1 = -3e^{2t} - 2e^{5t}\ , \qquad y_2 = 6e^{2t} + e^{5t}\ .$$

Ready to try one on your own?

Example 12. Solve the system $\dfrac{dy_1}{dt} = y_1 + y_2$, $\quad \dfrac{dy_2}{dt} = 4y_1 + y_2$.

Solution.

Eigenvalues and eigenvectors to the rescue of coupled differential equations.

Example 12, more writing space. Solve the system $\dfrac{dy_1}{dt} = y_1 + y_2$, $\quad \dfrac{dy_2}{dt} = 4y_1 + y_2$.

Maple syntax for this type of questions.

A Maple output for Exercise 12.

```
> with(LinearAlgebra):  # It won't work if you do not load the linear
  algrbra package. Try if you do not believe me.
> A:= <<1, 4>|<1,1>>;
```

$$A := \begin{bmatrix} 1 & 1 \\ 4 & 1 \end{bmatrix} \qquad (1)$$

```
> CharacteristicPolynomial(A, lambda); # using existing commands
```

$$\lambda^2 - 2\lambda - 3 \qquad (2)$$

(3)

```
> charPol := Determinant(A-t);    # using basic commands
  solve(charPol = 0, t);          # we find the eigenvalues
```

$$charPol := t^2 - 2t - 3$$

$$3, -1 \qquad (4)$$

```
> Eigenvectors(A); # gives the eigenvalues (first vector) and the
  corresponding eigenvectors (the columns of the matrix)
```

$$\begin{bmatrix} -1 \\ 3 \end{bmatrix}, \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} \\ 1 & 1 \end{bmatrix} \qquad (5)$$

TIP! *You can come back to this slide, or the course notes, or the 'First-Year Maple notes', which I cannot recommend enough, if you ever wonder which Maple commands to use to get the eigenvalues and the eigenvectors.*

What about *three* coupled first order ODEs?

Example 14. Solve the system of three differential equations in three unknowns

$$\frac{dy_1}{dt} = y_1 + 2y_2 - y_3 \ , \quad \frac{dy_2}{dt} = -6y_1 - 7y_2 + 3y_3 \ , \quad \frac{dy_3}{dt} = -4y_1 - 4y_2 + y_3 \ .$$

Solution. The coefficient matrix is $\begin{pmatrix} 1 & 2 & -1 \\ -6 & -7 & 3 \\ -4 & -4 & 1 \end{pmatrix}$. We have founds its eigenvalues and eigenvectors in Example 4 (pages 11-13):

$$E_{\lambda=-1} = \text{span}\left\{ \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} \right\} \quad \text{and} \quad E_{\lambda=-3} = \text{span}\left\{ \begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix} \right\}$$

Therefore, we can immediately write down the general solution of the system as

$$\vec{\mathbf{y}} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = Ce^{-t} \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + De^{-t} \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + Ee^{-3t} \begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix} \ .$$

Comment. The coefficient matrix in this case has a repeated eigenvalue $\lambda = -1$. From what has been learned in calculus about repeated roots, we might expect the solution to have a term involving e^{-t} and another involving te^{-t}. Note, however, that this is *not* the case – at least, not in this example. More work is done on this kind of problem in MATH2501 and MATH2601.

What about higher order ODEs?

Higher order equations. We can use the above methods to solve second (or higher) order differential equations in one unknown by converting them to a system of first order equations in two (or more) unknowns.

Example 15. Solve the 2nd order differential equation $\dfrac{d^2y}{dt^2} - 2\dfrac{dy}{dt} - 24y = 0$ (*)

Solution. Setting $y_1 = y$, $y_2 = \dfrac{dy_1}{dt}$ and $\overrightarrow{\mathbf{y}} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$, the equation (*) can be rewritten as

$$\frac{dy_1}{dt} = y_2\,, \qquad \frac{dy_2}{dt} = 24y_1 + 2y_2\,, \text{ i.e. } \begin{pmatrix} \frac{dy_1}{dt} \\ \frac{dy_2}{dt} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 24 & 2 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

or $\dfrac{d\overrightarrow{\mathbf{y}}}{dt} = \begin{pmatrix} 0 & 1 \\ 24 & 2 \end{pmatrix} \overrightarrow{\mathbf{y}}$. The coefficient matrix has eigenvalues and eigenvectors

$$\lambda = 6\,, \ \overrightarrow{\mathbf{v}} = \alpha \begin{pmatrix} 1 \\ 6 \end{pmatrix} \quad \text{and} \quad \lambda = -4\,, \ \overrightarrow{\mathbf{v}} = \alpha \begin{pmatrix} 1 \\ -4 \end{pmatrix}\,,$$

and so we obtain

$$\overrightarrow{\mathbf{y}} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = Ce^{6t} \begin{pmatrix} 1 \\ 6 \end{pmatrix} + De^{-4t} \begin{pmatrix} 1 \\ -4 \end{pmatrix}\,.$$

The solution of the original equation is $y = y_1$, that is, $y = Ce^{6t} + De^{-4t}$.

Reflection on the matrix method we used in the previous examples

If you are thinking

"So what? All we did was solve the 2nd order differential equation

$$\frac{d^2y}{dt^2} - 2\,\frac{dy}{dt} - 24y = 0$$

to find

$$y = Ce^{6t} + De^{-4t}$$

using a method more complicated than the one we learnt in Calculus.
So what is that eigenvalues/eigenvectors method good for??

...then keep in mind that the matrix/eigenvalues/eigenvectors method

- *also works for* ***higher*** *order differential equations, not just second order: From now on, you have no reason to be scared of a linear ODE with a* $y''' = y^{(3)}$ *or a* $y^{(4)}$ *in it;*
- *and, more importantly, it gives you a way to deal with* ***coupled*** *ODEs.*

Let's try a 3rd order ODE

Example 16.

Consider the 3rd order differential equation $y''' - 3y'' - 6y' + 8y = 0$. (*)

a) Rewrite (*) in the form $\dfrac{d\overrightarrow{\mathbf{y}}}{dt} = A\overrightarrow{\mathbf{y}}$, for a well-chosen matrix A and a well-chosen vector $\overrightarrow{\mathbf{y}}$. Note that $\overrightarrow{\mathbf{y}}$ will depend on t but A will not.

b) Using the Maple output provided on the next slide or otherwise (time permitting), solve (*).

Solution.

Example 16.
Consider the 3rd order differential equation $y''' - 3y'' - 6y' + 8y = 0$. (*)

a) Rewrite (*) in the form $\dfrac{d\overrightarrow{\mathbf{y}}}{dt} = A\overrightarrow{\mathbf{y}}$, for a well-chosen matrix A and a well-chosen vector $\overrightarrow{\mathbf{y}}$. Note that $\overrightarrow{\mathbf{y}}$ will depend on t but A will not.

b) Using the Maple output provided on the next slide or otherwise (time permitting), solve (*).

Solution.

$y_1 := y$

$y_2 := y' = y_1'$

$y_3 := y'' = y_2'$

(*) can be rewritten

$y''' = 3y'' + 6y' - 8y$

i.e.

$y_3' = 3y_3 + 6y_2 - 8y_1$
$= -8y_1 + 6y_2 + 3y_3$

Let $\vec{v} := \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}$

$$\therefore \frac{d\vec{v}}{dt} = \begin{pmatrix} y_1' \\ y_2' \\ y_3' \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -8 & 6 & 3 \end{pmatrix}}_{A} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}$$

Example 16.
Consider the 3rd order differential equation $y''' - 3y'' - 6y' + 8y = 0$. (*)

a) Rewrite (*) in the form $\dfrac{d\overrightarrow{\mathbf{y}}}{dt} = A\overrightarrow{\mathbf{y}}$, for a well-chosen matrix A and a well-chosen vector $\overrightarrow{\mathbf{y}}$. Note that $\overrightarrow{\mathbf{y}}$ will depend on t but A will not.

b) Using the Maple output provided on the next slide or otherwise (time permitting), solve (*).

Eigenvalues?

$$\chi(\lambda) = \det(A - \lambda I) = \det \begin{array}{c} + \\ - \\ + \end{array}\begin{pmatrix} -\lambda & 1 & 0 \\ 0 & -\lambda & 1 \\ -8 & 6 & 3-\lambda \end{pmatrix}$$

(expand along 1^{st} column)

$$= -\lambda \begin{vmatrix} -\lambda & 1 \\ 6 & 3-\lambda \end{vmatrix} + 0 - 8 \begin{vmatrix} 1 & 0 \\ -\lambda & 1 \end{vmatrix}$$

$$= -\lambda(\lambda^2 - 3\lambda - 6) - 8(1)$$

$$\boxed{\chi(\lambda) = -\lambda^3 + 3\lambda^2 + 6\lambda - 8 \quad \text{(Characteristic polynomial)}}$$

- We notice that $\chi(1) = 0$, so we can factor out $\lambda - 1$

$$\chi(\lambda) = -\lambda^3 + 3\lambda^2 + 6\lambda - 8 = (\lambda - 1)(-\lambda^2 + m\lambda + 8)$$

coeff of $\lambda^2 = 3 = m + 1 \Rightarrow m = 2$

$$\chi(\lambda) = (\lambda - 1)(-\lambda^2 + 2\lambda + 8)$$
$$= -(\lambda - 1)(\lambda^2 - 2\lambda - 8)$$
$$= -(\lambda - 1)(\lambda - 4)(\lambda + 2)$$

$\boxed{-8}$: -4, 2 $\qquad P = -8$, $S = -2$

The eigenvalues are the roots of $\chi(\lambda)$ ie

$$\boxed{\lambda = -2, \ \lambda = 1, \ \lambda = 4}$$

A Maple output for the matrix method in Exercise 16.

```
> # y''' -3 y'' - 6y' + 8y = 0, matrix method
  with(LinearAlgebra): # First, load the linear algebra package.
  A := <<0,0,-8>|<1,0,6>|<0,1,3>>;
```

$$A := \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -8 & 6 & 3 \end{bmatrix} \qquad (6)$$

```
> CharPol:= CharacteristicPolynomial(A, lambda);
```

$$CharPol := \lambda^3 - 3\lambda^2 - 6\lambda + 8 \qquad (7)$$

```
> solve(CharPol = 0, lambda);
```

$$1, -2, 4 \qquad (8)$$

```
> Eigenvectors(A); # gives the eigenvalues (first vector) and the
  corresponding eigenvectors (the columns of the matrix)
```

$$\begin{bmatrix} -2 \\ 1 \\ 4 \end{bmatrix}, \begin{bmatrix} \frac{1}{4} & 1 & \frac{1}{16} \\ -\frac{1}{2} & 1 & \frac{1}{4} \\ 1 & 1 & 1 \end{bmatrix} \qquad (9)$$

Let's try a 3rd order ODE, continued

Checking with Maple, for Exercise 16, using 'dsolve'.

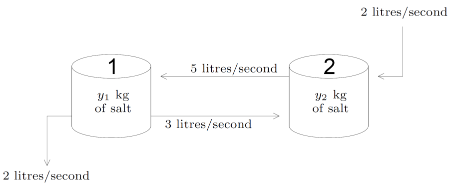
```
> # y''' -3 y'' - 6y' + 8y = 0, Calculus
  ODE := diff(y(x), x,x,x) - 3*diff(y(x),x,x) - 6*diff(y(x),x)+8*y(x)=
  0;
  dsolve(ODE);
```

$$ODE := \frac{d^3}{dx^3}\,y(x) - 3\,\frac{d^2}{dx^2}\,y(x) - 6\,\frac{d}{dx}\,y(x) + 8\,y(x) = 0$$

$$y(x) = \_C1\,e^{4x} + \_C2\,e^{-2x} + \_C3\,e^{x} \qquad (10)$$

Back to the Salt example

To conclude, we return to the salt–in–tanks problem (Example 10 page 30),



using `Maple` for the calculations.

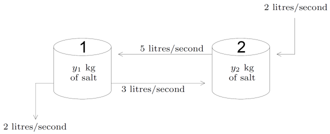
```
> with(LinearAlgebra):
  A := <<-0.05,0.03>|<0.05,-0.05>>:
  Eigenvectors(A);
```

$$\begin{bmatrix} -0.0112701665379258 + 0.\,I \\ -0.0887298334620742 + 0.\,I \end{bmatrix}, \begin{bmatrix} 0.790569415042095 + 0.\,I & -0.790569415042095 + 0.\,I \\ 0.612372435695794 + 0.\,I & 0.612372435695794 + 0.\,I \end{bmatrix} \qquad \textbf{(12)}$$

The output indicates that A has an eigenvalue $\lambda = -0.0113$ with eigenvectors any multiple of $(0.791, 0.612)^T$, and similarly for the other eigenvalue.
So the general solution of the equation is

$$\vec{\mathbf{y}} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = C_1 e^{-0.0113t} \begin{pmatrix} 0.791 \\ 0.612 \end{pmatrix} + C_2 e^{-0.0887t} \begin{pmatrix} -0.791 \\ 0.612 \end{pmatrix}.$$

....*continued*



Entering this into `Maple` and then using the following commands to take into account the initial conditions.

```
> y := C1*exp(-0.01113*t)*<0.791, 0.612>+ C2*exp(-0.0887*t)*<-0.791, 0.612>;
  y0 := subs(t=0,y);
  solve( {y0[1]=4,y0[2]=7}, {C1,C2} );
  subs(%,y);
```

$$y := \begin{bmatrix} 0.791\, C1\, e^{-0.01113\, t} - 0.791\, C2\, e^{-0.0887\, t} \\ 0.612\, C1\, e^{-0.01113\, t} + 0.612\, C2\, e^{-0.0887\, t} \end{bmatrix}$$

$$y0 := \begin{bmatrix} 0.791\, C1\, e^{-0.} - 0.791\, C2\, e^{-0.} \\ 0.612\, C1\, e^{-0.} + 0.612\, C2\, e^{-0.} \end{bmatrix}$$

$$\{C1 = 8.247399255,\ C2 = 3.190509242\}$$

$$\begin{bmatrix} 6.523692811\, e^{-0.01113\, t} - 2.523692810\, e^{-0.0887\, t} \\ 5.047408344\, e^{-0.01113\, t} + 1.952591656\, e^{-0.0887\, t} \end{bmatrix}$$

gives the final answer

$$\vec{y} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 6.52e^{-0.0113t} - 2.52e^{-0.0887t} \\ 5.05e^{-0.0113t} + 1.95e^{-0.0887t} \end{pmatrix}$$

....continued

The final answer is

$$\vec{\mathbf{y}} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 6.52e^{-0.0113t} - 2.52e^{-0.0887t} \\ 5.05e^{-0.0113t} + 1.95e^{-0.0887t} \end{pmatrix}$$

Plot of the amount of salt over the first 5 minutes.

```
> C1 := 8.247399255: C2 := 3.190509242:
  plot( {y[1],y[2]}, t=0..300, view=0..7);
```